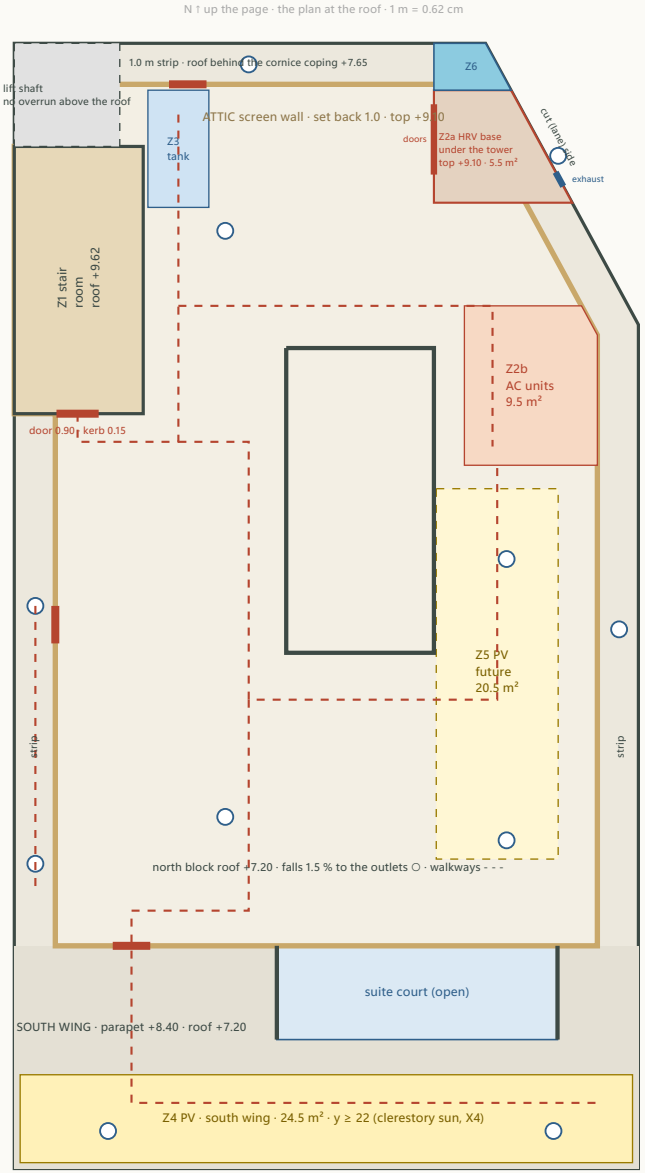
Roof study · the stair to the roof, the set-back attic, the main cornice, the reserved zones · Taif / Palm House V7.1

Design review F-07 (major) · owner decisions 31 Oct 2026 · flat roof, no tiles · the roof for maintenance and equipment only · no slab level changes · all sizes assumed until the engineer and the suppliers confirm



1 · Roof plan: what is on the roof

- **Access:** the main stair's two extra flights arrive in the **stair room Z1** on the well (x 0–2.75, y 2.2–7.9); its door (0.90, opening out, on a 0.15 m kerb) opens south onto the roof. The lift stops at the first floor; its shaft needs no head above the roof (§ 5).
- **Attic:** a screen wall 1.8 m high (top +9.10 with its coping) set back 1 m from the facade face on the north, the cut (lane) and the east sides, and on the west except where the stair room stands on the facade line; its south wall stands on the y 19.25 line (beams B8 / R5). Length ≈ 52 m. Two doors in it to the strip (north, west) and one to the south wing.
- **The 1.0 m strip** (Z7) in front of the attic: roof behind the main cornice and its low coping, with its own outlets.
- **South wing** (y 19.25–24): the main cornice and a plain parapet to +8.40 the roofline steps 0.7 m down from the attic coping, and the parapet stands on the facade while the attic is 1.0 m back.
- **Guards:** a black steel bar guard 1.10 m round the courtyard sky hole (19.3 m) and on the suite court's west and east edges; its south edge keeps the gym roof railing (X4).
- **Drains:** 11 roof outlets to internal downpipes in the walls (no scuppers through the cornice); falls 1.5 % away from the holes and the attic walls; the strip falls along the facade to its outlets; overflow spouts in the strip's upstand.
- **Walkways:** 0.9 m wide on paving pads from the stair room door to every zone, the holes' guards, the strip doors and the south wing.

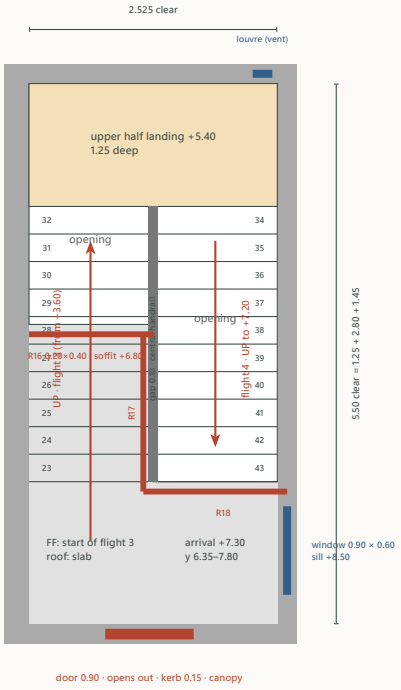
Zone	m²	Note
Z1 stair room (footprint)	15.7	on the well; inside 2.525 × 5.50; roof +9.62
Z2a the HRV base under the tower	5.5	`studies/air_system.typ` v3.1 (owner): the tower rises from a base housing the ventilation unit (one mass): top +9.10 = the attic coping, 1.65 m clear inside (a low service compartment: the unit ≈ 110 kg on a plinth right behind a pair of doors 1.5 m high on its west face, filters and the core pulled from the roof terrace, no entry); the intake duct drops inside the tower's compartment; the exhaust louvre on the cut face
Z2b plant: AC outdoor units	9.5	behind the cut / east attic wall, SOUTH of the tower and its base (owner): 2.2 m clear of the base's south wall, 5.15 m south of the intake louvre in plan, so the tower never takes their hot air; over the suite bath (anti-vibration mounts); 0.6 m service space round each unit
Z3 water tank(s)	3.3	≈ 2 × 1.5 m³ between the roof beams R10 and R3, inside the attic (needed only if the owner wants a gravity reserve)
Z4 PV, south wing (now reserved)	24.5	y ≥ 21.95: no shade on the clerestories (X4); fixing points cast in before waterproofing
Z5 PV, future (north block east)	20.5	inside the attic; morning shade from the east attic wall
Z6 malqaf head	1.4	the tower rises to +9.80: the one element above the attic, by design; v3: a four-sided head, the breeze shaft and the ventilation intake compartment inside (`studies/air_system.typ`)
Z7 the 1.0 m strip in front of the attic	35.6	roof behind the main cornice and its coping; falls to its own outlets; reached through two doors in the attic wall

Areas on wall centre lines / zone outlines. PV: ≈ 12.2 + 10.3 panels of 2 m² (≈ 9.5 kWp at 420 W each) if both zones are used.

The stair to the roof · plan at the roof slab and section through the well · same well, same geometry: 22 risers × 0.16 = 3.60 m, goings 0.28

lift shaft below (y 0–2.2): no head above the roof

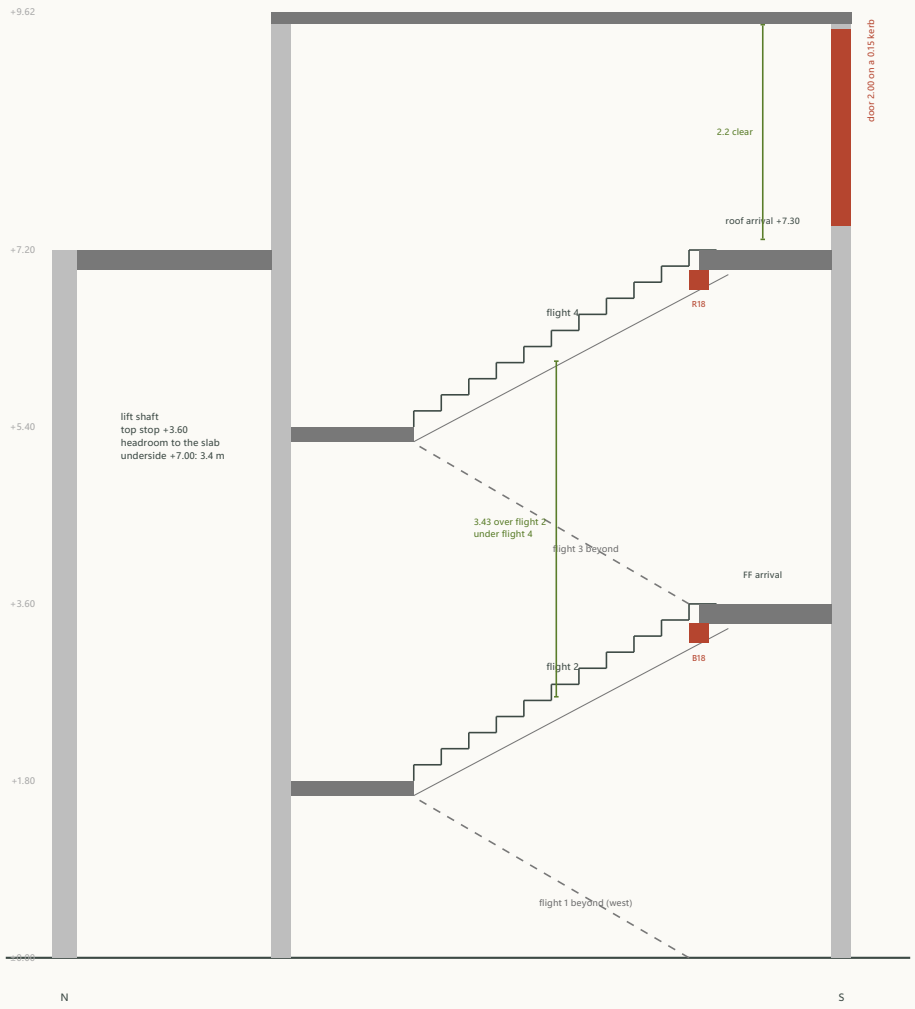
stair room plan at the roof slab · white = opening · grey = slab · north up



door 0.90 · opens out · kerb 0.15 · canopy

section through flights 2 and 4 (x ≈ 2.0), looking west · flights 1 and 3 beyond dashed · R16 over flight 3: headroom 2.22 at tread 6

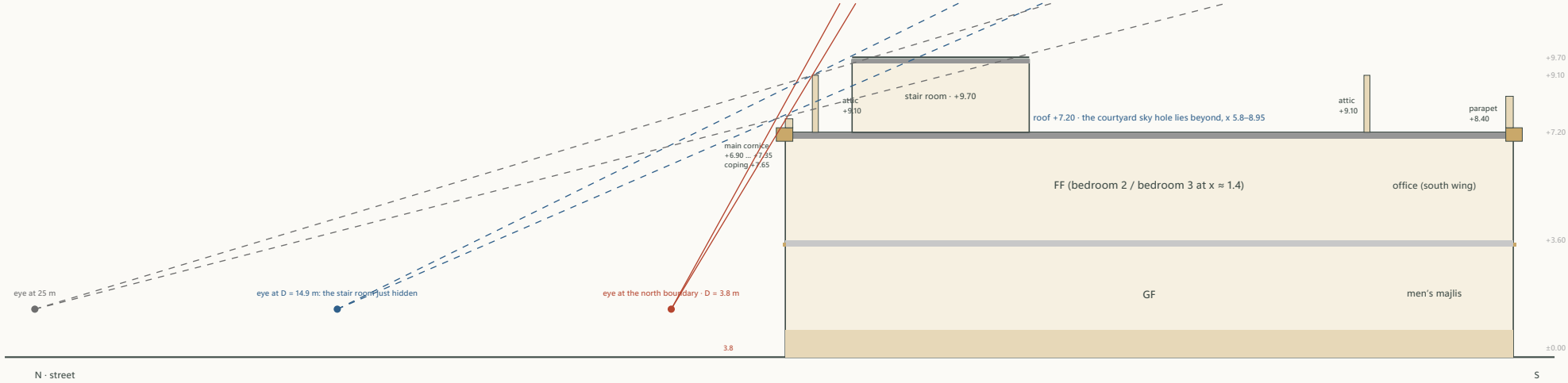
stair room roof · soffit +9.50 · top +9.62 (slab 0.12, assumed)



2 · The stair, one storey up

- **Flights 3 and 4** repeat flights 1 and 2: 11 risers each × 0.16, goings 0.28 flight 3 (west) up from the first-floor arrival strip (y 6.35–7.80, +3.60) to the upper half landing (y 2.30–3.55, +5.40); flight 4 (east) up to the roof arrival (+7.30) on y 6.35–7.80. Flight widths 1.21 m, the 0.10 gap with the central handrail; wall handrails on the west wall (flight 3) and on flight 4's east edge a balustrade (its east side is the well, open to the first-floor gallery): **handrails on both sides of every flight**.
- **Roof opening:** the same L as the first floor, one storey up: over flight 4 and the half landing (x 1.39–2.65, y 2.30–6.35), over flight 3 beyond tread 6 (x 0.125–1.39, y 2.30–4.75). Trimmers **R16 / R17 / R18** 0.20 × 0.40 total, soffit +6.80, laid out as B16 / B17 / B18 (R16 and R18 bear on R17, R17 on R16 and R18's ends on the walls / R10). Headroom over flight 3 at tread 6: **2.22 m** to R16 (as 2.22 to B16); over flight 2 under flight 4: 3.43 m; over flight 4 to the stair room soffit ≥ 2.3 m.
- **Stair room:** on the well's walls (x 0 outer 250, x 2.75, y 2.2, y 7.9: 200 mm block, plastered cream, honey stone plinth outside), inside **2.525 × 5.50 m**; floor = the roof slab round the opening (the arrival y 6.35–7.80 and the strip x 0.125–1.29, y 4.85–6.35); roof slab 0.12 (or an insulated steel deck) with its soffit at **+9.50** = **2.2 m clear** over the arrival; top +9.62, edge +9.70. Door 0.90 × 2.00 in the south wall opening out, on a 0.15 m kerb (head +9.45) with a drip canopy, weather-stripped, a brush seal and a storm sill (rain and dust); a 0.90 × 0.60 window in the east wall (sill +8.50) and a louvre high in the north wall for light and cross-ventilation; a light and a socket.
- **First floor:** nothing changes: the rooms, the landing (y 7.9–9.2), the gallery and the family living stay; the first-floor arrival strip becomes the foot of flight 3 as in any multi-storey stair; the floor strip x 0.125–1.29, y 4.85–6.35 (already inside the stair box, not a room) lies under flight 3's first treads (a closed triangle). The x 2.75 edge of the well stays open to the gallery with its railing.
- **Lift:** no change to the shaft (2.25 × 2.2, placeholder) and no overrun above the roof: see § 5.

Sight lines · north–south section through the stair room (x ≈ 1.4), looking west · the attic 1.8 m set back 1 m · eye 1.6 m above the ground



3 · What the attic hides, and from where

The eye is 1.6 m above the ground outside (+1.30 against the house's ±0.00; the ground assumed at −0.30). An object behind the facade is hidden when its top lies below the line from the eye over the screen in front of it (the cornice coping +7.65 on the facade, or the attic coping +9.10 1 m behind it).

Observer (north street)	cornice	attic	stair room
north boundary (street), NW corner	59.1°	58.4° (hidden)	54.3° hidden
north street, 10 m from the facade	32.4°	35.3°	34.5° hidden
north street, the limit	23°	26.1°	26.1° hidden
north street, 25 m (the far side)	14.3°	16.7°	17.1° **shows**

Angles above the horizon from the eye. The stair room's north face is 2.23 m behind the north facade face, its top +9.70. From the lane boundary (D = 2.5 m) the AC units (1.2 m, 2.0 m behind the cut facade) are seen at 57.6° against the attic's 65.8°: hidden; and from any distance, since 1.2 < 1.8. From the east boundary (D = 3.6 m) the plant zone is likewise hidden. The malqaf head (+9.80, on the facade in the cut tip) is the one element above the attic, by design.

- **Attic height 1.8 m:** with 1.5 m the stair room would show above the attic from 9.2 m on; with 1.8 m from **14.9 m** (≈ 11.1 m beyond the plot boundary, across the street). Beyond that the stair room shows 0.6 m above the attic line: a small corner element over the north-west corner, 2.2 m back from the north facade. The stair room cannot be lower (2.2 m clear over the arrival) and cannot be set back on the west: the well stands on the west wall.
- **The one conflict with the brief:** the stair room is "inside the attic" in plan but 0.6 m above its coping, and on the west it stands on the facade line (the attic is set back there everywhere else). To hide it from every distance the attic would need ≈ 2.5 m on the north and west: outside the 1.5–1.8 m the owner gave. **Recommendation: accept** (hidden from the boundary and from across the street; the west faces the neighbour at 0.785 m).

4 · The courtyard's sky and sun with the attic

From every point of the courtyard floor the attic walls and the stair room lie **below the hole's own edges** (the roof edge beams at +7.20), so they take nothing from the sky the courtyard sees today (enclosure study: 7.2 % of the hemisphere from the floor's middle, unchanged):

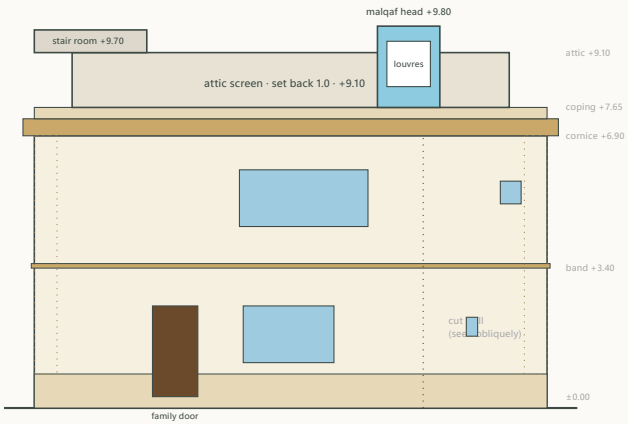
From the far edge of the floor	hole edge	attic / stair room
attic south wall (y 19.25, +9.10) from the floor's north edge (y 6.6)	48.4°	35.7° ✓
attic north wall (y 0.875) from the floor's south edge (y 12.9)	48.4°	37.1° ✓
stair room (x 0–2.75, +9.70) from the floor's east edge (x 8.85)	67.9°	57.8° ✓
attic east wall (x 12.436) from the floor's west edge (x 4.35)	58°	48.4° ✓

- **Winter noon sun** (45.2°, the light check): the ray that just clears the hole's south edge (+7.20 at y 13.0) passes the attic's south wall line (y 19.25) at +13.49, far above its coping +9.10: the attic never shades the courtyard, in any season. The clerestory windows sit in the y 19.25 wall itself, below the attic wall: unchanged; their sun comes over the gym roof edge (X4), which stays a railing.
- **The suite court** gets the attic's south wall on its north side (its shaded side): no loss of sun, a little less sky.

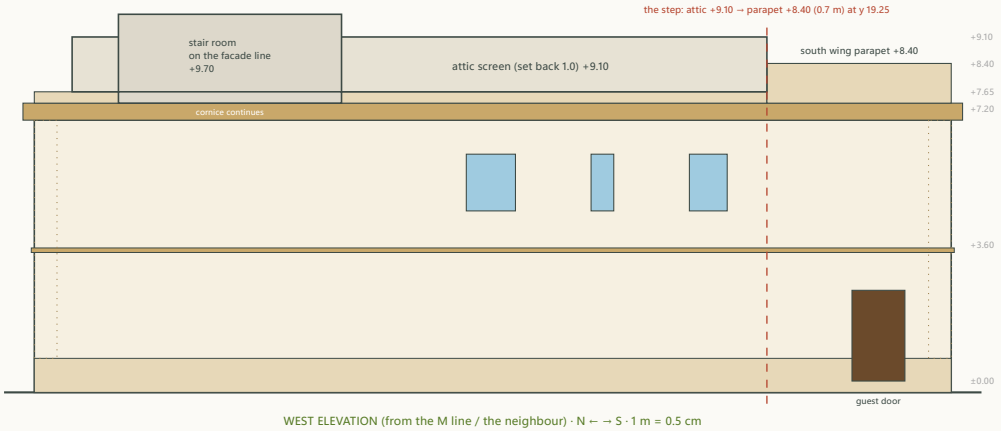
5 · The lift

- Top stop at the first floor (+3.60); the roof slab underside at +7.00: **3.4 m headroom available**. A typical Saudi villa lift (machine-room-less traction, 400–630 kg, 2 stops) asks 3.4–3.6 m headroom above the top floor and a 1.2–1.4 m pit (assumed from supplier catalogues; open item 14 asks for the drawings); a platform home lift (EN 81-41) asks 2.3–2.6 m.
- **So:** choose a lift with headroom ≤ 3.4 m: no shaft head above the roof. If the chosen lift needs up to 3.85 m, a raised shaft head of ≤ 0.45 m (top +7.65) stays hidden behind the main cornice coping; more than that is not possible: the shaft (x 0–2.25, y 0–2.2) lies in the 1.0 m strip in front of the attic, not inside it. **No conflict** with the stair room (it starts at y 2.2, the shaft's south wall). Added to open item 14 as a requirement for the supplier.

North and west elevations · simple line drawings · the main cornice, the thin cornice between the floors, the set-back attic, the stepped roofline · openings indicative



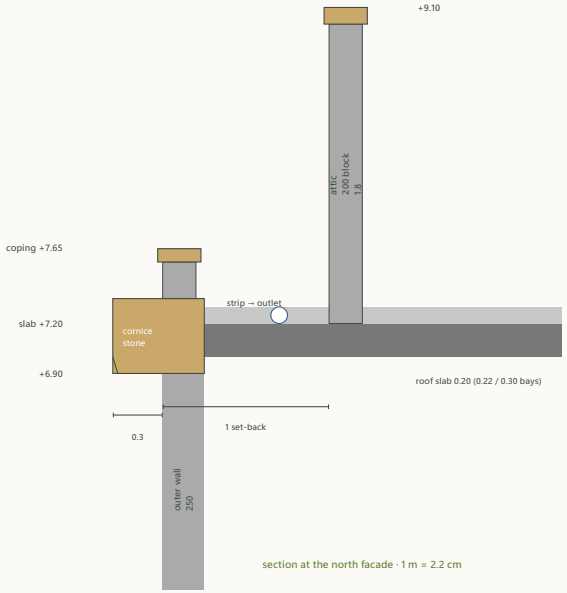
NORTH ELEVATION (from the street) · W ← → E · 1 m = 0.5 cm



WEST ELEVATION (from the M line / the neighbour) · N ← → S · 1 m = 0.5 cm

Cream plaster, honey stone at the plinth (to +0.60), the corners (quoins, dotted), the thin band at the first-floor slab (0.12 m, projecting 0.08) and the main cornice (0.45 m high, projecting 0.3, +6.90 … +7.35); the attic and the stair room plastered cream with stone copings. The south elevation repeats the west one's south part (parapet +8.40); the east elevation the north block's coping and attic (set back 1.0) over the kitchen / suite windows, then the parapet. Openings are indicative here: the real ones are in the plans and the studies.

The cornice and the attic in section · loads for the engineer · what changes in the files



6 · Sizes (assumed, by this study)

- **Main cornice:** honey stone, 0.45 m high (+6.90 … +7.35), projecting 0.3 m with a drip, on the facade edge all round the house; above it a 0.30 m upstand with a stone coping to **+7.65** on the north block (the strip's parapet) and the plain parapet to **+8.40** on the south wing. The thin cornice between the floors: a stone band 0.12 m high at +3.40, projecting 0.08.
- **Attic:** 200 mm block, plastered cream, stone coping; 1.8 m above the roof slab (coping +9.10); set back 1 m from the facade face (0.88 m from the wall centre lines); ≈ 52 m long; on the south on the y 19.25 line (B8 / R5). Three doors (two to the strip, one to the south wing).
- **Roofline step:** 0.7 m at y 19.25 (attic coping +9.10 → parapet +8.40), plus the 1.0 m plan set-back.
- **Stair room:** 2.75 × 5.70 on walls (2.525 × 5.50 inside), roof top +9.62, 2.2 m clear; 2.5 m above the roof slab, 0.6 m above the attic coping.

7 · Loads for the engineer (assumed)

- **Attic walls:** 200 block 1.8 m + coping ≈ 4.5 kN/m, standing on the roof slab 0.88 m inside the outer-wall beams (R7 / R8, the cut and east wall beams): near the supports, but a line load on the slab: thicken the slab edge or add a small hidden beam under it; the south attic wall on R5 / B8.
- **Stair room:** its own roof slab 0.12 (≈ 3.0 kN/m² + 1.5 live) on four block walls on the well's walls; the trimmers R16–R18 as B16–B18.
- **Cornice:** stone ≈ 2.5 kN/m hung on the slab edge (anchored precast or cut stone); parapets 1.1 m ≈ 2.8 kN/m; the thin band 0.3 kN/m.
- **Plant:** 3–4 AC outdoor units ≈ 1.0 kN each and the ERV ≈ 1.5 kN on pads with anti-vibration mounts (local ≈ 1.5 kN/m²); **tank(s):** up to 3 m³ = 30 kN on ≈ 3.3 m² (≈ 9 kN/m²) on the R10–R3 bay (1.5 m span): to be checked; **PV:** ≈ 0.25 kN/m² plus wind uplift (ballast or cast-in fixing points, 24 m² now, 20 m² future); **malqaf head** as the air study; walkway pads 0.5 kN/m².
- **Roof live load** for maintenance only: 1.0–1.5 kN/m² (SBC 301, the engineer).

8 · Conflicts and open points

- **Stair room vs attic height** (§ 3): the stair room's roof is 0.6 m above the attic coping and stands on the west facade line; hidden from the north boundary and up to 14.9 m; the owner accepts, or raises the attic to ≈ 2.5 m on the north and west.
- **Lift** (§ 5): no overrun if the supplier's headroom is ≤ 3.4 m (≤ 3.85 m with a 0.30 m raised head behind the cornice): a requirement for open item 14.
- **Courtyard light** (§ 4): none lost.
- **First-floor layout** (§ 2): nothing lost.
- **3D and the plans:** the roof opening, the stair room, the attic, the cornice and the parapets are not yet in `3d/params.yaml` / `build_3d.py` (the builder has no attic or cornice yet) nor in the presentation: PENDING_CHANGES.md.

9 · What this study changes in the files

- `studies/structure.typ`: R16 / R17 / R18 in `beams-roof`, the stair well in `voids-roof`, notes X16 (stair, trimmers, stair room) and X17 (attic, parapets, cornice, plant loads); the stair page's note on the roof.
- `docs/DESIGN_LANGUAGE.md`: flat roof, no tiles, the main cornice, the set-back attic on the north block, the south wing parapet.
- `data/decisions.csv`: roof access by the main stair; the roofline. Open item 14: the lift headroom requirement (`docs/PENDING_CHANGES.md`).

Study checks passed (12)